

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (Biotechnology) (Credit System) Degree Examinations
May - June 2023

20BTC104 - CANCER BIOLOGY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five** full questions choosing **Two** full questions from **Unit – I** & **Unit – III** each and **One** full question from **Unit – II**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	List the characteristics features of cell lines.	10	L*2	1	1.2.1
b)	What is chronic myelogenous leukemia? Discuss how changes in chromosomal structure and number is responsible for it.	10	L2	2	1.2.1
2. a)	Estrogen exerts its effects contributing to breast cancer. Justify the statement.	10	L2	1	1.2.1
b)	How colon tumor progression takes place? Add a note on drug resistance in cancer cells.	10	L2	2	1.2.1
3. a)	How DNA tumor viruses induce cancer? Explain.	10	L2	1	1.2.1
b)	Taking retinoblastoma as an example, explain how changes in the structure of individual chromosome results in cancer.	10	L2	2	1.2.1
Unit – II					
4. a)	Growth factor gene can become an oncogene. Justify appropriately giving an example.	10	L2	3	1.2.1
b)	Depict Ras signaling pathway involvement in tumorigenesis.	10	L2	3	1.2.1
5. a)	Discuss the role of telomere and telomerase in cancer.	10	L2	3	1.2.1
b)	List important growth factors for angiogenesis. How VEGF is regulated by HIF-1 to induce angiogenesis.	10	L2	3	1.2.1
Unit – III					
6. a)	Discuss how cyclin-CDK complexes are modulated in order to impose control on specific steps in the cell cycle through CDK inhibitors.	10	L2	4	1.2.1
b)	How cytokines are helpful in immune therapy? Explain.	10	L2	5	1.4.1
7. a)	How CDK's respond to mitogenic and antimitogenic signals? Explain.	10	L2	4	1.2.1
b)	Give an account on important plant secondary metabolites functioning as chemotherapeutic drugs.	10	L2	5	1.4.1
8. a)	Discuss on Neurofibromatosis (NF1) functions in tumorigenesis.	10	L2	4	1.2.1
b)	Comment on therapeutic applications of monoclonal antibodies in the treatment of cancer.	10	L2	5	1.4.1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
